Міністерство освіти і науки України
Національний технічний університет України
«Київський політехнічний інститут імені Ігоря Сікорського»
Факультет інформатики та обчислювальної техніки
Кафедра обчислювальної техніки

Лабораторна робота №3

з дисципліни
«Об’єктно-орієнтоване програмування»

на тему

«Розробка інтерфейсу користувача»

Виконала:

студентка групи ІМ-52

Семенюк Ліліана Валентинівна

Номер у списку - 21

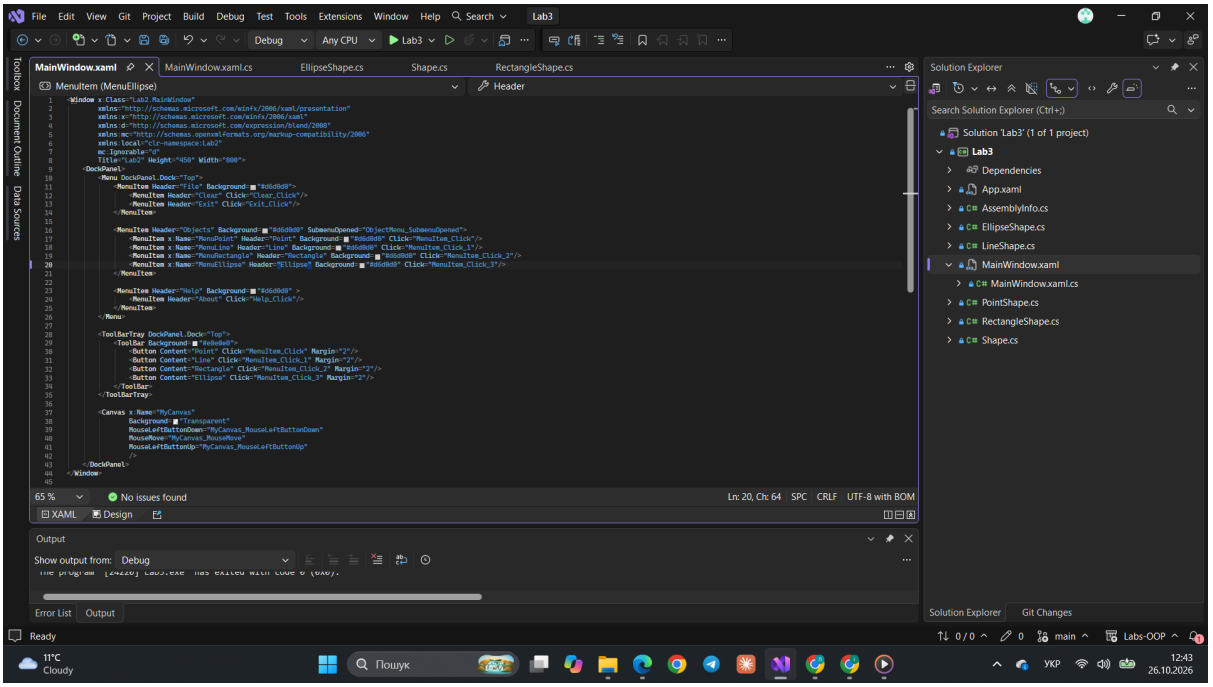
Київ 2026

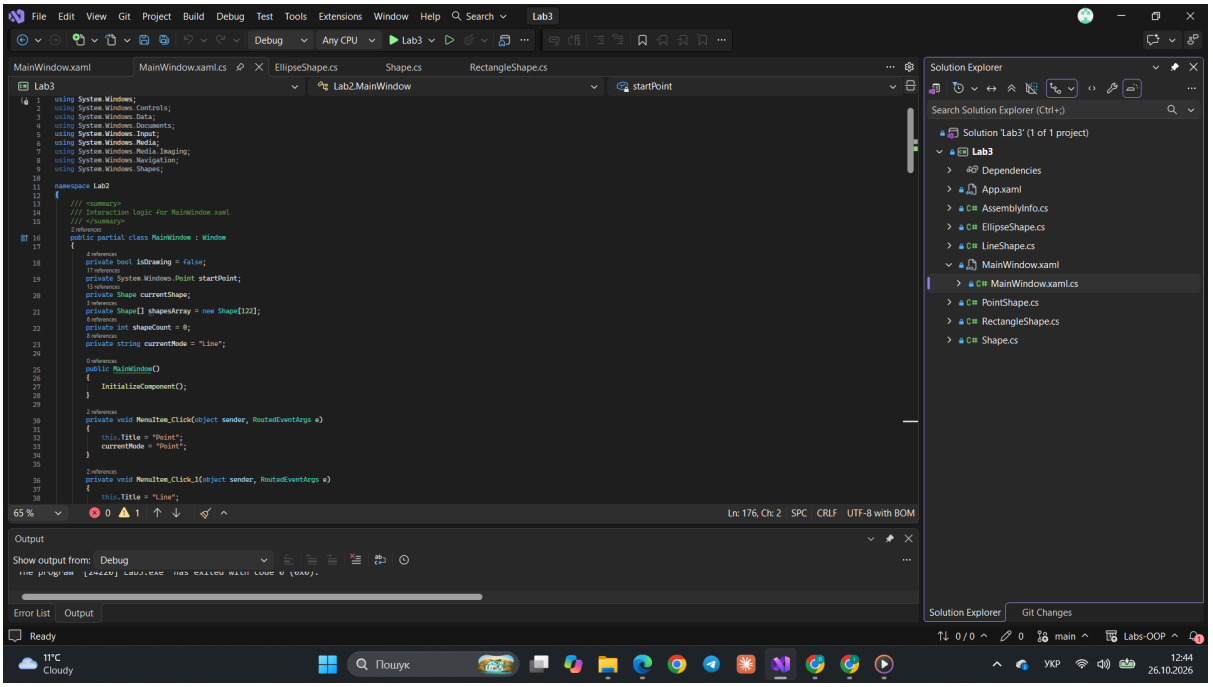
Завдання:
1. Створити у середовищі розробки проект з ім’ям Lab3.
2. Написати вихідний текст програми згідно варіанту завдання.
3. Скомпілювати вихідний текст і отримати виконуваний файл програми.
4. Перевірити роботу програми. За потреби налагодити програму.
5. Проаналізувати та прокоментувати результати та вихідний текст програми.
6. Оформити звіт.

Завдання за варіантом:

1. Написати статичний масив з 122 елементами.
2. Зробити “гумовий ” слід синього кольору.
3. Увід прямокутника по двох протилежних кутах.
4. Колір заповнення прямокутника рожевий.
5. Ввід еліпса від центру до зовнішніх кутів
6. Колір заповнення еліпса білий
7. Додати позначку поточного типу об’єкту, що вводиться в меню

Код:


File Edit View Git Project Build Debug Test Tools Extensions Window Help Search Lab3
Debug Any CPU Lab3
MainWindow.xaml MainWindow.xaml.cs EllipseShape.cs Shape.cs RectangleShape.cs
MenuItem (MenuEllipse) Header
<Window x:Class="Lab2.MainWindow"
xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml"
xmlns:d="http://schemas.microsoft.com/expression/blend/2008"
xmlns:mc="http://schemas.openxmlformats.org/markup-compatibility/2006"
xmlns:local="clr-namespace:Lab2"
mc:Ignorable="d"
Title="Lab2" Height="450" Width="800">
<DockPanel>
<Menu DockPanel.Dock="Top">
<MenuItem Header="File" Background="#d6d9d0">
<MenuItem Header="Clear" Click="Clear_Click"/>
<MenuItem Header="Exit" Click="Exit_Click"/>
</MenuItem>
<MenuItem Header="Objects" Background="#d6d9d0" SubmenuOpened="ObjectMenu_SubmenuOpened">
<MenuItem x:Name="MenuPoint" Header="Point" Background="#d6d9d0" Click="MenuItem_Click"/>
<MenuItem x:Name="MenuLine" Header="Line" Background="#d6d9d0" Click="MenuItem_Click_1"/>
<MenuItem x:Name="MenuRectangle" Header="Rectangle" Background="#d6d9d0" Click="MenuItem_Click_2"/>
<MenuItem x:Name="MenuEllipse" Header="Ellipse" Background="#d6d9d0" Click="MenuItem_Click_3"/>
</MenuItem>
<MenuItem Header="Help" Background="#d6d9d0" >
<MenuItem Header="About" Click="Help_Click"/>
</MenuItem>
</Menu>
<ToolBarTray DockPanel.Dock="Top">
<ToolBar Background="#e6e9d0">
<Button Content="Point" Click="MenuItem_Click" Margin="2"/>
<Button Content="Line" Click="MenuItem_Click_1" Margin="2"/>
<Button Content="Rectangle" Click="MenuItem_Click_2" Margin="2"/>
<Button Content="Ellipse" Click="MenuItem_Click_3" Margin="2"/>
</ToolBar>
</ToolBarTray>
<Canvas x:Name="MyCanvas"
Background="Transparent"
MouseLeftButtonDown="MyCanvas_MouseLeftButtonDown"
MouseMove="MyCanvas_MouseMove"
MouseLeftButtonUp="MyCanvas_MouseLeftButtonUp"
/>
</DockPanel>
</Window>
65 % No issues found Ln: 20, Ch: 64 SPC CRLF UTF-8 with BOM
XAML Design
Output
Show output from: Debug
Error List Output
Solution Explorer
Search Solution Explorer (Ctrl+;)
Solution 'Lab3' (1 of 1 project)
Lab3
Dependencies
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LineShape.cs
MainWindow.xaml
MainWindow.xaml.cs
PointShape.cs
RectangleShape.cs
Shape.cs
Solution Explorer Git Changes
Ready
Labs-OOP
11°C Cloudy
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УКР
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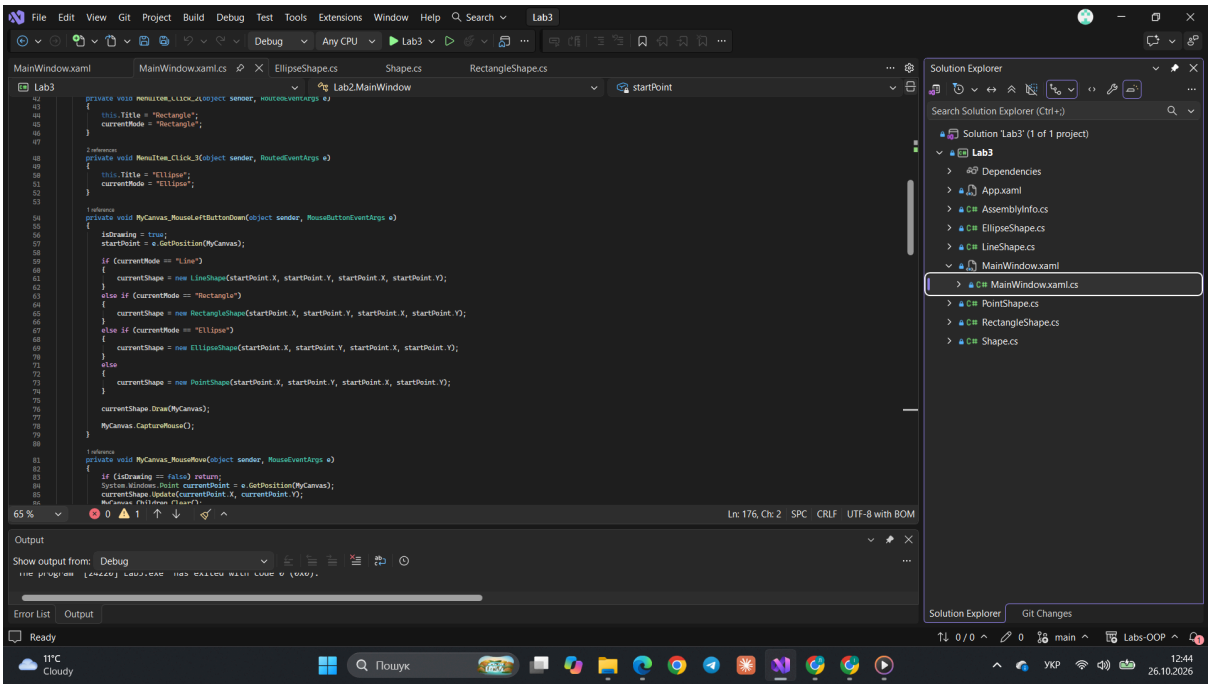
```
using System.Windows;
using System.Windows.Controls;
using System.Windows.Data;
using System.Windows.Documents;
using System.Windows.Input;
using System.Windows.Media;
using System.Windows.Media.Imaging;
using System.Windows.Navigation;
using System.Windows.Shapes;

namespace Lab2
{
    /// <summary>
    /// Interaction logic for MainWindow.xaml
    /// </summary>
    public partial class MainWindow : Window
    {
        private bool isDrawing = false;
        private System.Windows.Point startPoint;
        private Shape currentShape;
        private Shape[] shapesArray = new Shape[122];
        private int shapeCount = 0;
        private string currentMode = "Line";

        public MainWindow()
        {
            InitializeComponent();
        }

        private void MenuItem_Click(object sender, RoutedEventArgs e)
        {
            this.Title = "Point";
            currentMode = "Point";
        }

        private void MenuItem_Click_1(object sender, RoutedEventArgs e)
        {
            this.Title = "Line";
```



```
        private void MenuItem_Click_2(object sender, RoutedEventArgs e)
        {
            this.Title = "Rectangle";
            currentMode = "Rectangle";
        }

        private void MenuItem_Click_3(object sender, RoutedEventArgs e)
        {
            this.Title = "Ellipse";
            currentMode = "Ellipse";
        }

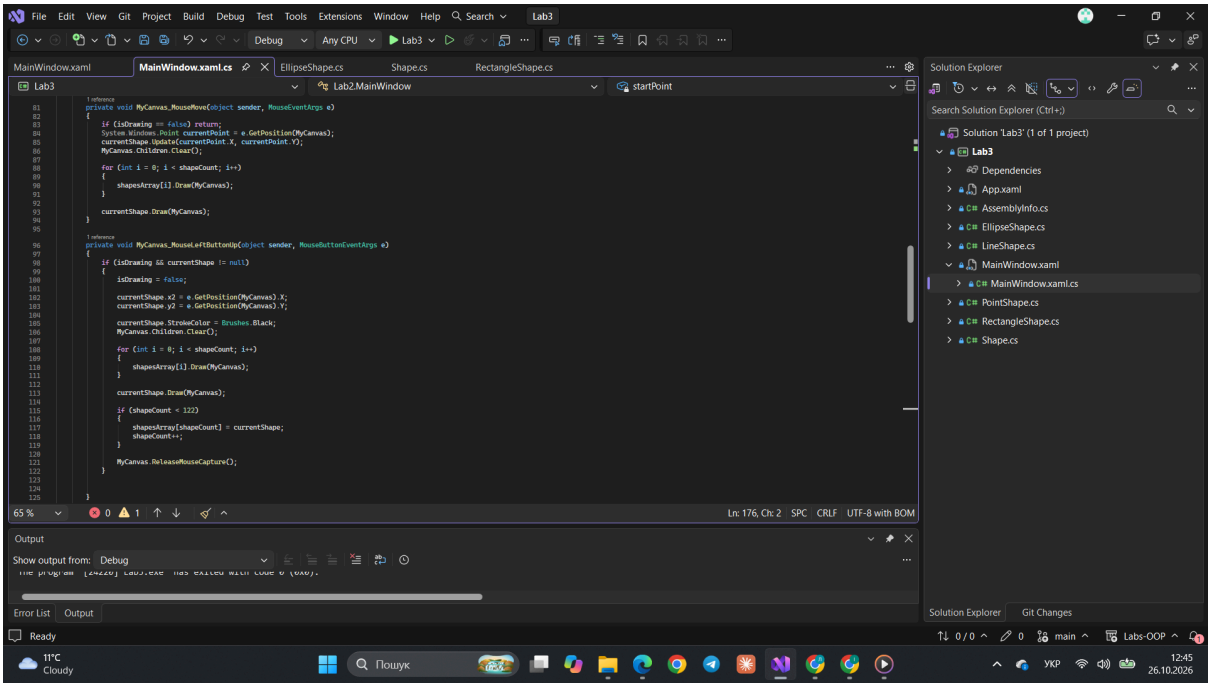
        private void MyCanvas_MouseLeftButtonDown(object sender, MouseButtonEventArgs e)
        {
            isDrawing = true;
            startPoint = e.GetPosition(MyCanvas);

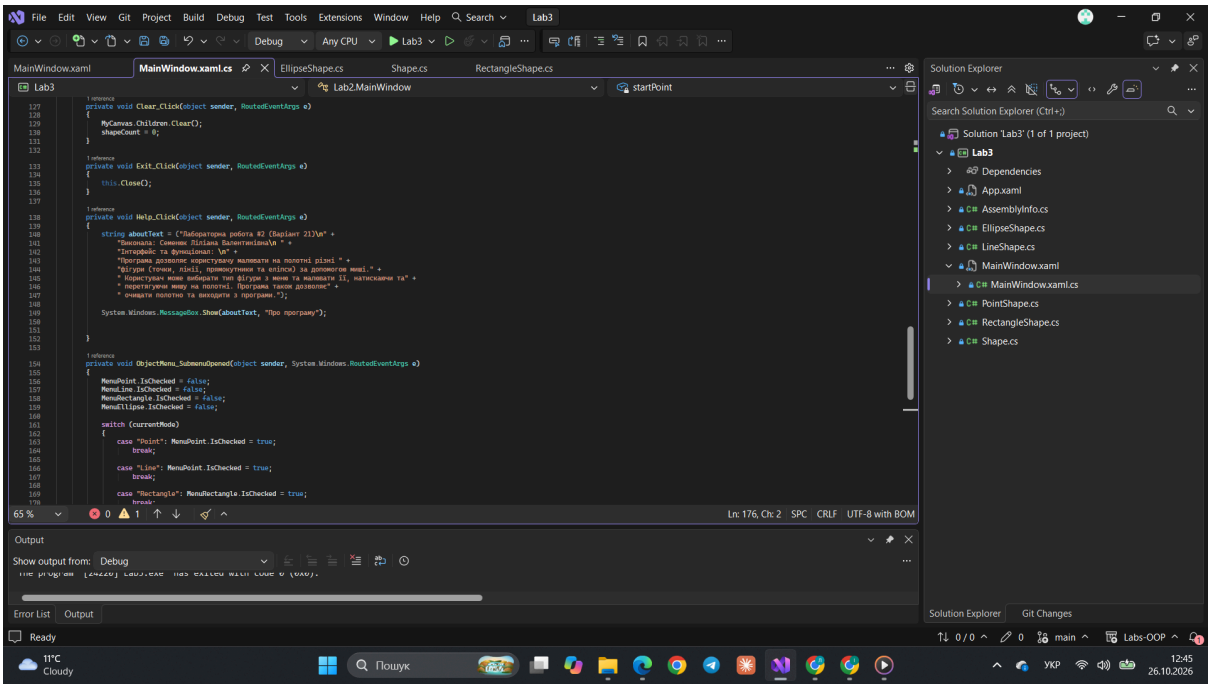
            if (currentMode == "Line")
            {
                currentShape = new LineShape(startPoint.X, startPoint.Y, startPoint.X, startPoint.Y);
            }
            else if (currentMode == "Rectangle")
            {
                currentShape = new RectangleShape(startPoint.X, startPoint.Y, startPoint.X, startPoint.Y);
            }
            else if (currentMode == "Ellipse")
            {
                currentShape = new EllipseShape(startPoint.X, startPoint.Y, startPoint.X, startPoint.Y);
            }
            else
            {
                currentShape = new PointShape(startPoint.X, startPoint.Y, startPoint.X, startPoint.Y);
            }

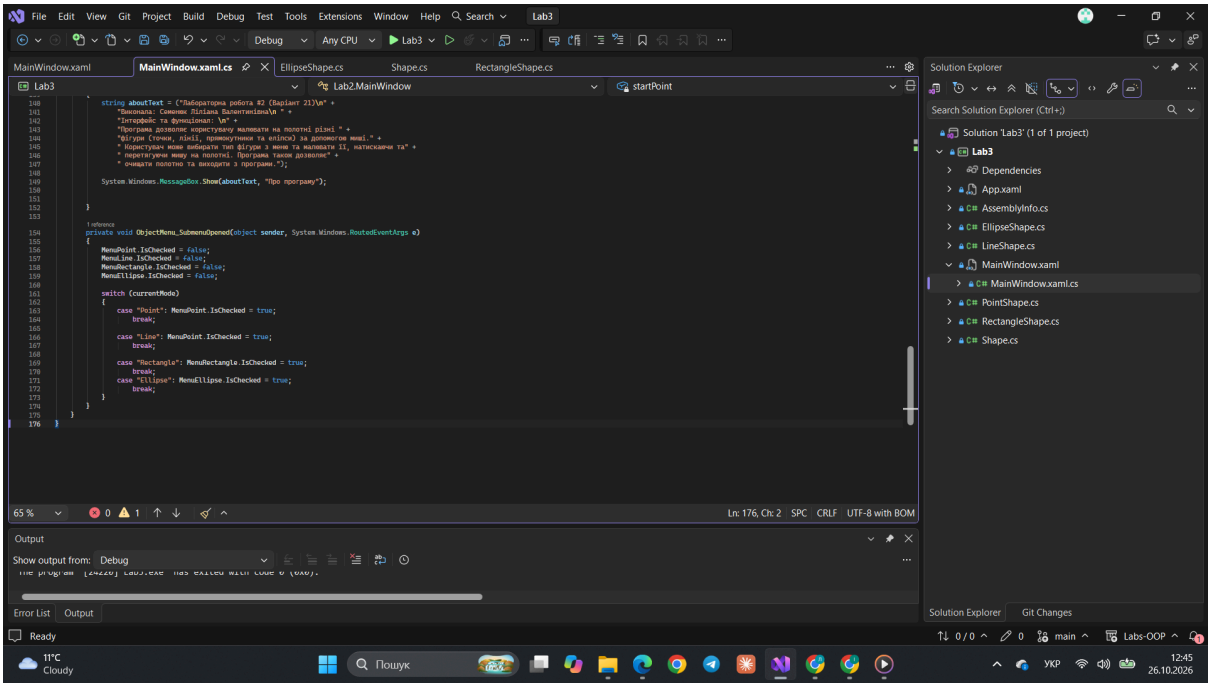
            currentShape.Draw(MyCanvas);

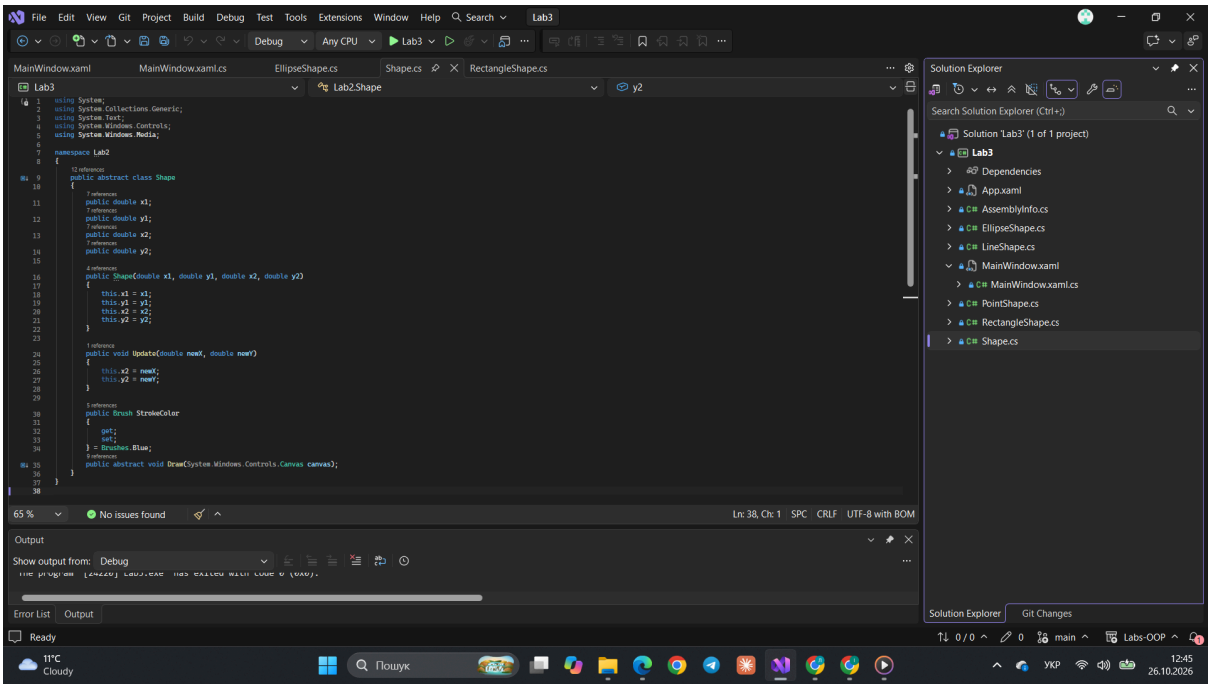
            MyCanvas.CaptureMouse();
        }

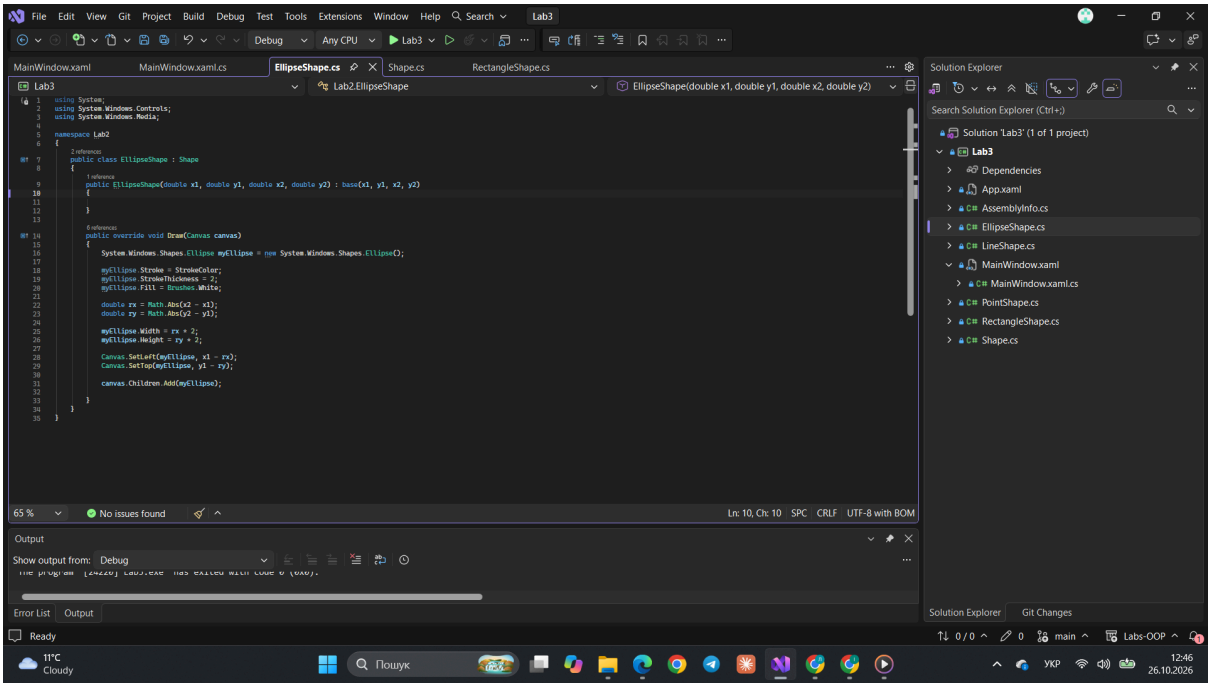
        private void MyCanvas_MouseMove(object sender, MouseEventArgs e)
        {
            if (isDrawing == false) return;
            System.Windows.Point currentPoint = e.GetPosition(MyCanvas);
            currentShape.Update(currentPoint.X, currentPoint.Y);
```

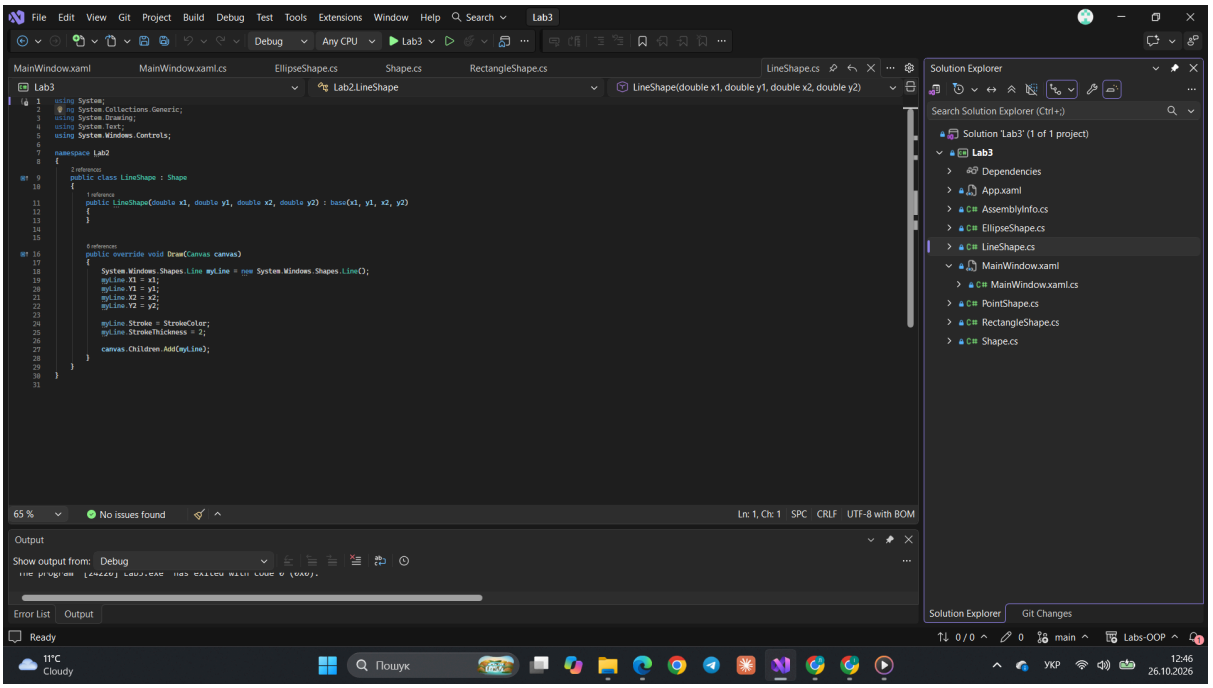

File Edit View Git Project Build Debug Test Tools Extensions Window Help Search Lab3
MainWindow.xaml
MainWindow.xaml.cs
EllipseShape.cs
Shape.cs
RectangleShape.cs
Lab3
Lab2.MainWindow
startPoint
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Solution 'Lab3' (1 of 1 project)
Lab3
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RectangleShape.cs
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Output
Show output from: Debug
Error List
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Ready
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Labs-OOP
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УКР
12:45
26.10.2026

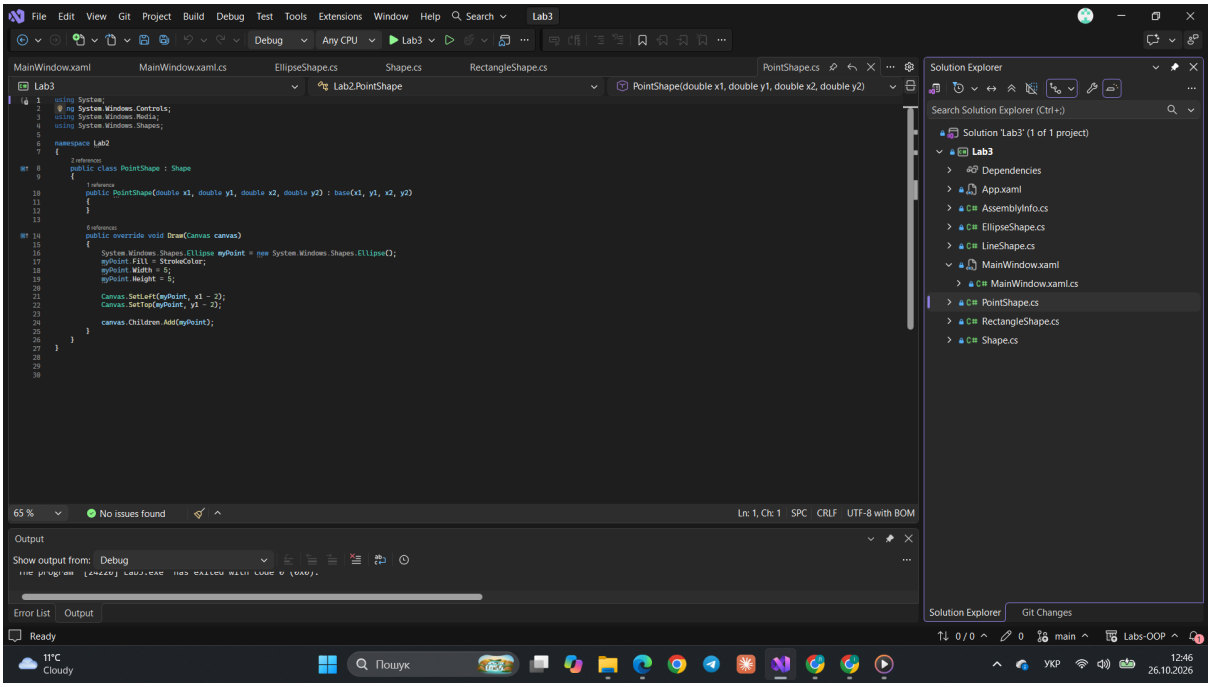

File Edit View Git Project Build Debug Test Tools Extensions Window Help Search Lab3
MainWindow.xaml
MainWindow.xaml.cs
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Shape.cs
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Lab2.MainWindow
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12:45
26.10.2026

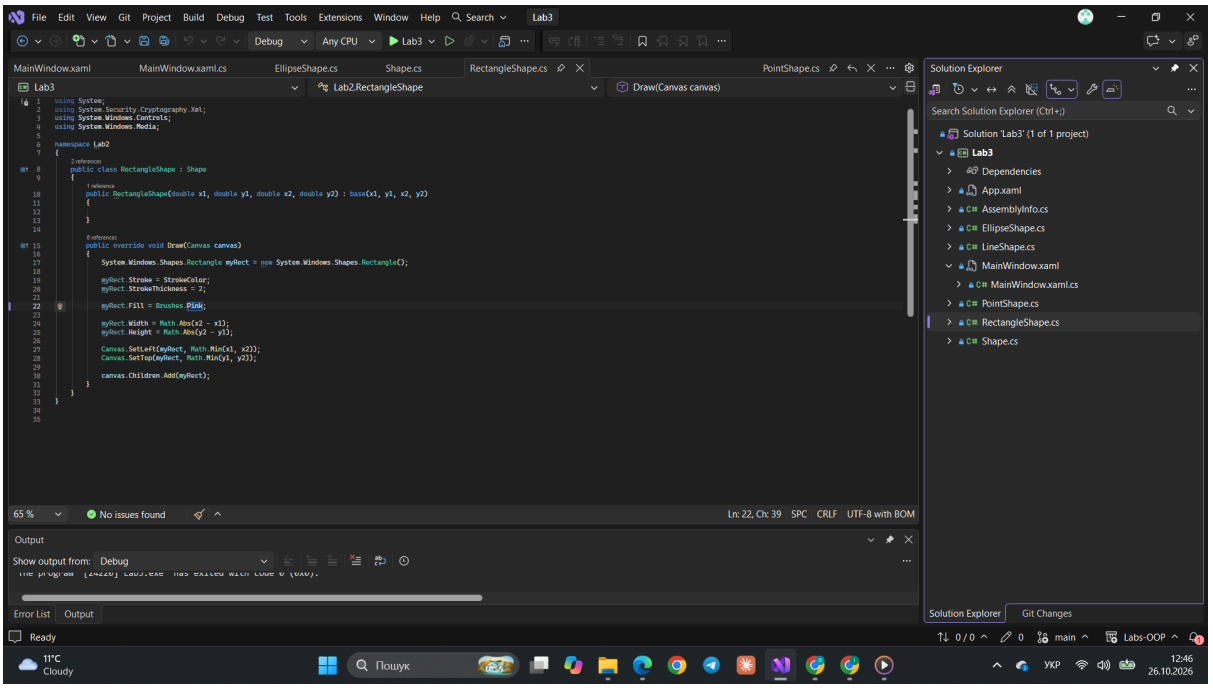

File Edit View Git Project Build Debug Test Tools Extensions Window Help Search Lab3
MainWindow.xaml
MainWindow.xaml.cs
EllipseShape.cs
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RectangleShape.cs
Lab3
Lab2.MainWindow
startPoint
string aboutText = ("Лабораторна робота #2 (Варіант 21)\n" +
"Виконала: Семенюк Ліліана Валентинівна\n " +
"Інтерфейс та функціонал: \n" +
"Програма дозволяє користувачу малювати на полотні різні " +
"фігури (точки, лінії, прямокутники та еліпси) за допомогою миші." +
" * Користувач може вибирати тип фігури з меню та малювати її, натискаючи та" +
" * перетягуючи мишу на полотні. Програма також дозволяє" +
" * очищати полотно та виходити з програми.");
System.Windows.MessageBox.Show(aboutText, "Про програму");
1 reference
private void ObjectMenu_SubmenuOpened(object sender, System.Windows.RoutedEventArgs e)
{
MenuPoint.IsChecked = false;
MenuLine.IsChecked = false;
MenuRectangle.IsChecked = false;
MenuEllipse.IsChecked = false;
switch (currentMode)
{
case "Point": MenuPoint.IsChecked = true;
break;
case "Line": MenuPoint.IsChecked = true;
break;
case "Rectangle": MenuRectangle.IsChecked = true;
break;
case "Ellipse": MenuEllipse.IsChecked = true;
break;
}
}
}
65 %
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Output
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Solution 'Lab3' (1 of 1 project)
Lab3
Dependencies
App.xaml
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MainWindow.xaml.cs
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File Edit View Git Project Build Debug Test Tools Extensions Window Help Search Lab3
MainWindow.xaml
MainWindow.xaml.cs
EllipseShape.cs
Shape.cs
RectangleShape.cs
Lab3
Lab2.Shape
y2
using System;
using System.Collections.Generic;
using System.Text;
using System.Windows.Controls;
using System.Windows.Media;
namespace Lab2
{
12 references
public abstract class Shape
{
7 references
public double x1;
7 references
public double y1;
7 references
public double x2;
7 references
public double y2;
4 references
public Shape(double x1, double y1, double x2, double y2)
{
this.x1 = x1;
this.y1 = y1;
this.x2 = x2;
this.y2 = y2;
}
1 reference
public void Update(double newX, double newY)
{
this.x2 = newX;
this.y2 = newY;
}
5 references
public Brush StrokeColor
{
get;
set;
} = Brushes.Blue;
8 references
public abstract void Draw(System.Windows.Controls.Canvas canvas);
}
}
65 %
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Output
Show output from: Debug
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Search Solution Explorer (Ctrl+;)
Solution 'Lab3' (1 of 1 project)
Lab3
Dependencies
App.xaml
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Solution Explorer Git Changes
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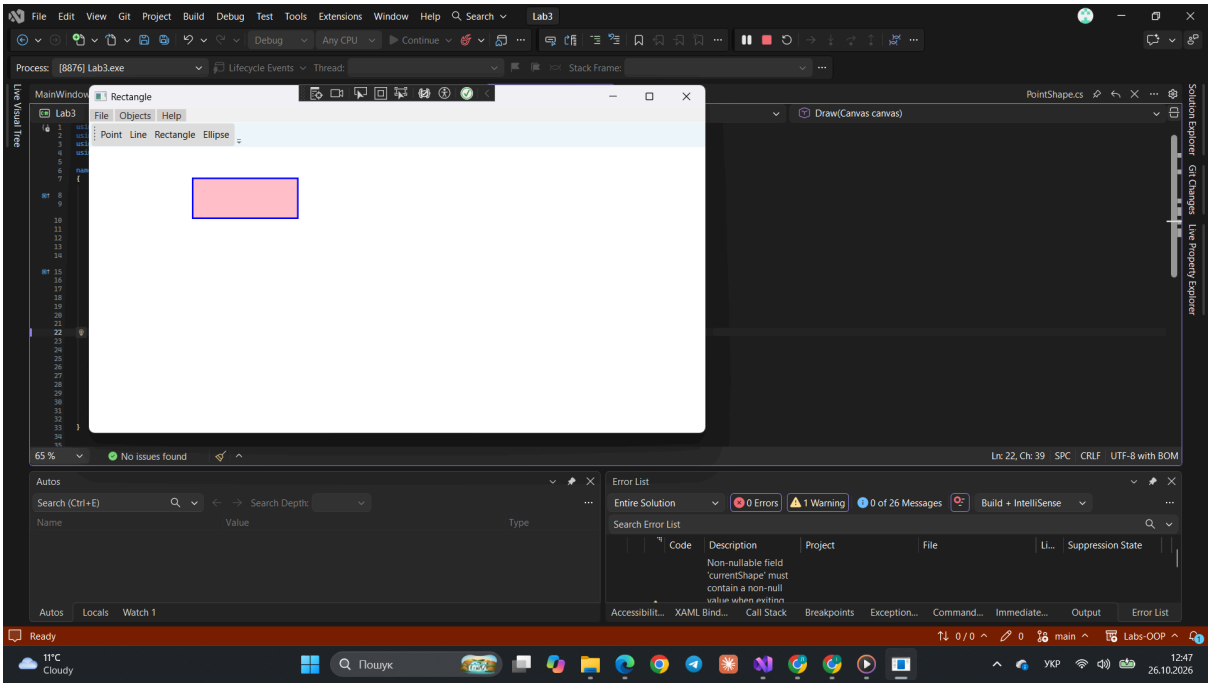

File Edit View Git Project Build Debug Test Tools Extensions Window Help Search Lab3
MainWindow.xaml
MainWindow.xaml.cs
EllipseShape.cs
Shape.cs
RectangleShape.cs
Lab3
Lab2.EllipseShape
EllipseShape(double x1, double y1, double x2, double y2)
using System;
using System.Windows.Controls;
using System.Windows.Media;
namespace Lab2
{
2 references
public class EllipseShape : Shape
{
1 reference
public EllipseShape(double x1, double y1, double x2, double y2) : base(x1, y1, x2, y2)
{
}
0 references
public override void Draw(Canvas canvas)
{
System.Windows.Shapes.Ellipse myEllipse = new System.Windows.Shapes.Ellipse();
myEllipse.Stroke = StrokeColor;
myEllipse.StrokeThickness = 2;
myEllipse.Fill = Brushes.White;
double rx = Math.Abs(x2 - x1);
double ry = Math.Abs(y2 - y1);
myEllipse.Width = rx * 2;
myEllipse.Height = ry * 2;
Canvas.SetLeft(myEllipse, x1 - rx);
Canvas.SetTop(myEllipse, y1 - ry);
canvas.Children.Add(myEllipse);
}
}
}
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Output
Show output from: Debug
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Solution Explorer
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Solution 'Lab3' (1 of 1 project)
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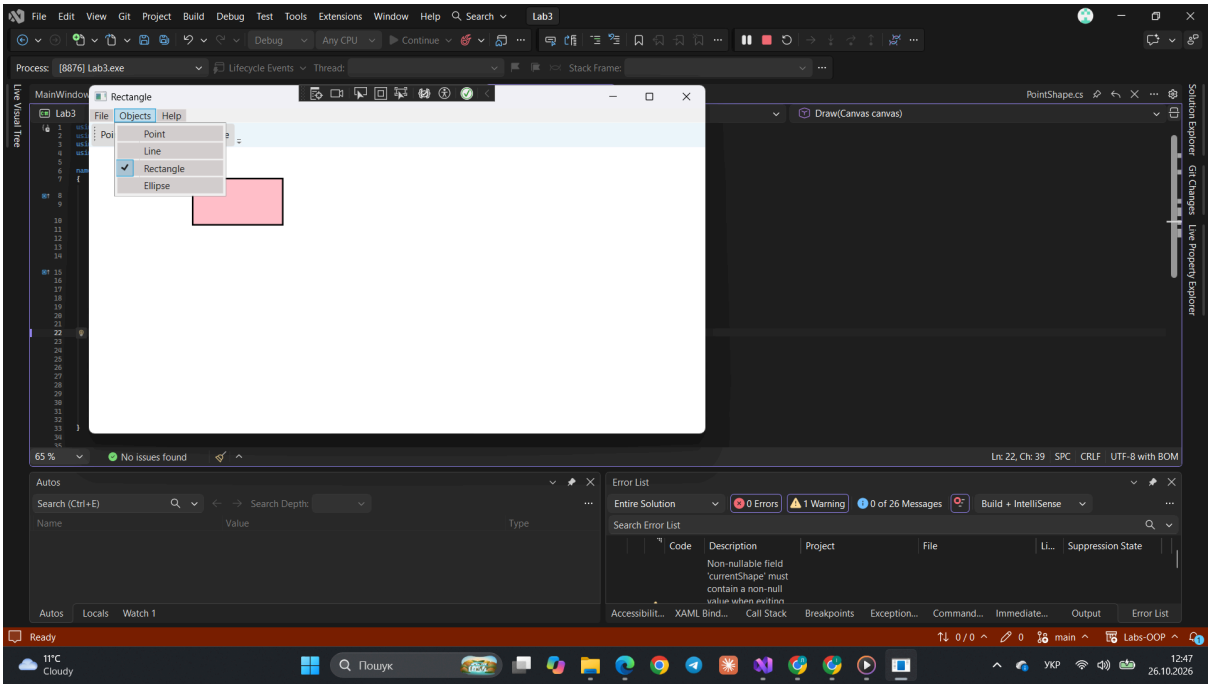

File Edit View Git Project Build Debug Test Tools Extensions Window Help Search Lab3
MainWindow.xaml
MainWindow.xaml.cs
EllipseShape.cs
Shape.cs
RectangleShape.cs
LineShape.cs
Lab3
Lab2.LineShape
LineShape(double x1, double y1, double x2, double y2)
using System;
using System.Collections.Generic;
using System.Drawing;
using System.Text;
using System.Windows.Controls;
namespace Lab2
{
2 references
public class LineShape : Shape
{
1 reference
public LineShape(double x1, double y1, double x2, double y2) : base(x1, y1, x2, y2)
{
}
0 references
public override void Draw(Canvas canvas)
{
System.Windows.Shapes.Line myLine = new System.Windows.Shapes.Line();
myLine.X1 = x1;
myLine.Y1 = y1;
myLine.X2 = x2;
myLine.Y2 = y2;
myLine.Stroke = StrokeColor;
myLine.StrokeThickness = 2;
canvas.Children.Add(myLine);
}
}
}
65 %
No issues found
Ln: 1, Ch: 1
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Output
Show output from: Debug
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Output
Solution Explorer
Search Solution Explorer (Ctrl+;)
Solution 'Lab3' (1 of 1 project)
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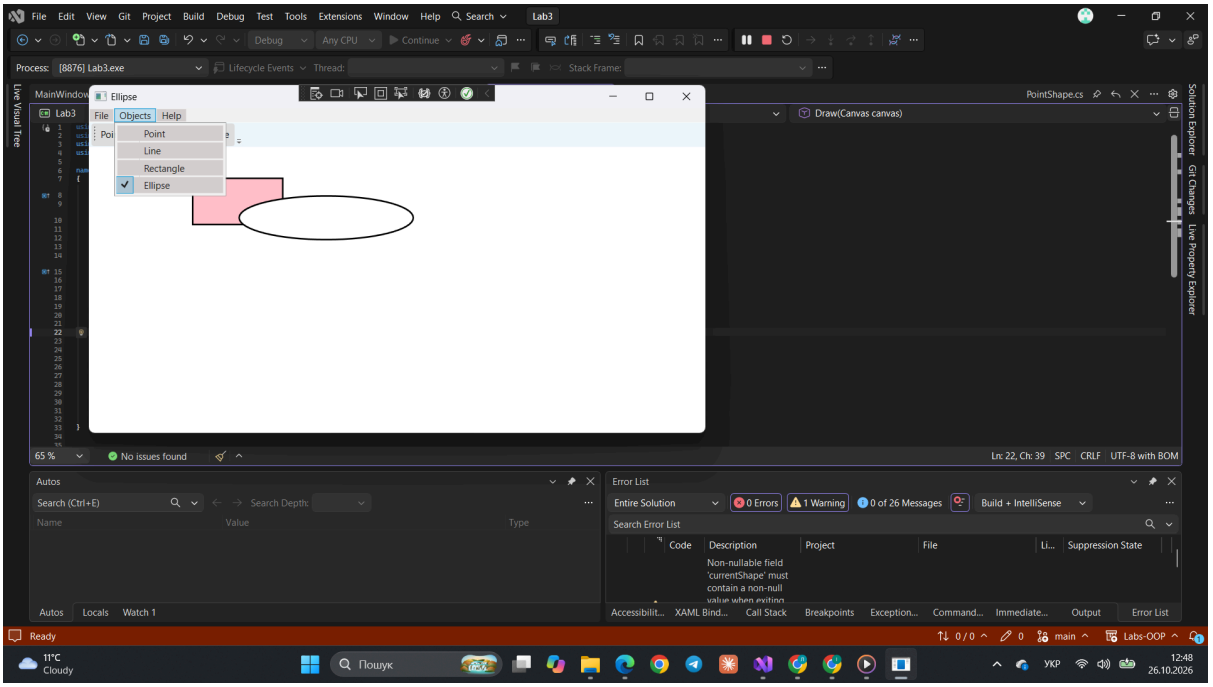

File Edit View Git Project Build Debug Test Tools Extensions Window Help Search Lab3
MainWindow.xaml MainWindow.xaml.cs EllipseShape.cs Shape.cs RectangleShape.cs PointShape.cs
Lab3 Lab2.PointShape PointShape(double x1, double y1, double x2, double y2)
using System;
using System.Windows.Controls;
using System.Windows.Media;
using System.Windows.Shapes;
namespace Lab2
{
2 references
public class PointShape : Shape
{
1 reference
public PointShape(double x1, double y1, double x2, double y2) : base(x1, y1, x2, y2)
{
}
0 references
public override void Draw(Canvas canvas)
{
System.Windows.Shapes.Ellipse myPoint = new System.Windows.Shapes.Ellipse();
myPoint.Fill = StrokeColor;
myPoint.Width = 5;
myPoint.Height = 5;
Canvas.SetLeft(myPoint, x1 - 2);
Canvas.SetTop(myPoint, y1 - 2);
canvas.Children.Add(myPoint);
}
}
}
65 % No issues found Ln: 1, Ch: 1 SPC CRLF UTF-8 with BOM
Output
Show output from: Debug
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Solution 'Lab3' (1 of 1 project)
Lab3
Dependencies
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MainWindow.xaml.cs
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Shape.cs
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File Edit View Git Project Build Debug Test Tools Extensions Window Help Search Lab3
MainWindow.xaml MainWindow.xaml.cs EllipseShape.cs Shape.cs RectangleShape.cs PointShape.cs
Lab3 Lab2.RectangleShape Draw(Canvas canvas)
using System;
using System.Security.Cryptography.Xml;
using System.Windows.Controls;
using System.Windows.Media;
namespace Lab2
{
2 references
public class RectangleShape : Shape
{
1 reference
public RectangleShape(double x1, double y1, double x2, double y2) : base(x1, y1, x2, y2)
{
}
0 references
public override void Draw(Canvas canvas)
{
System.Windows.Shapes.Rectangle myRect = new System.Windows.Shapes.Rectangle();
myRect.Stroke = StrokeColor;
myRect.StrokeThickness = 2;
myRect.Fill = Brushes.Pink;
myRect.Width = Math.Abs(x2 - x1);
myRect.Height = Math.Abs(y2 - y1);
Canvas.SetLeft(myRect, Math.Min(x1, x2));
Canvas.SetTop(myRect, Math.Min(y1, y2));
canvas.Children.Add(myRect);
}
}
}
65 % No issues found Ln: 22, Ch: 39 SPC CRLF UTF-8 with BOM
Output
Show output from: Debug
Error List Output
Solution Explorer
Search Solution Explorer (Ctrl+;)
Solution 'Lab3' (1 of 1 project)
Lab3
Dependencies
App.xaml
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LineShape.cs
MainWindow.xaml
MainWindow.xaml.cs
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RectangleShape.cs
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Результат:


Rectangle
File Objects Help
Point Line Rectangle Ellipse
Draw(Canvas canvas)
No issues found
Error List
Entire Solution
0 Errors
1 Warning
0 of 26 Messages
Build + IntelliSense
Non-nullable field 'currentShape' must contain a non-null value when exiting
Ready


Rectangle
File Objects Help
Point
Line
Rectangle
Ellipse
Draw(Canvas canvas)
No issues found
Error List
Entire Solution
0 Errors
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Висновок.

У ході виконання лабораторної роботи №3 за варіантом №22 було вдосконалено графічний редактор на WPF.

У програмі динамічний список фігур замінено на статичний масив із 122 елементів із керуванням через лічильник. Також змінено математику побудови еліпса: тепер точка кліку стає його центром, а рух миші визначає радіуси.

Інтерфейс доповнено панеллю інструментів із кнопками та оновленим меню. Завдяки події відкриття підменю програма автоматично позначає галочкою поточний обраний інструмент перед розгортанням списку.

Усі поставлені завдання виконано, програма працює коректно.